CHAPTER

29

Getting Started with DSPs

Once you decide that a Digital Signal Processor is right for your application, you need a way to get started. Many manufacturers will sell you a low cost evaluation kit, allowing you to experience their products first-hand. These are a great educational tool; it doesn't matter if you are a novice or a pro, they are the best way to become familiar with a particular DSP. For instance, Analog Devices provides the EZ-KIT® Lite to teach potential customers about its SHARC® family of Digital Signal Processors. For only $179, you receive all the hardware and software you need to see the DSP in action. This includes "canned" programs provided with the kit, as well as applications you can write yourself in assembly or C. Suppose you buy one of these kits from Analog Devices and play with it for a few days. This chapter is an overview of what you can expect to find and learn.

The ADSP-2106x family

In the last chapter we looked at the general operation of the ADSP-2106x "SHARC" family of Digital Signal Processors. Table 29-1 shows the various members of this family. All these devices use the same architecture, but have different amounts of on-chip memory, a key factor in deciding which one to use. Memory access is a common bottleneck in DSP systems. The SHARC DSPs address this by providing an ample supply of on-chip dual-ported SRAM. However, the last thing you want to do is pay for more memory than you need. DSPs often go into cost sensitive products, such as cellular telephones and CD players. In other words, the organization of this family is determined by *marketing* as well as *technology*.

The oldest member of this family is the ADSP-21020. This chip contains the core architecture, but does not include on-chip memory or I/O handling. This means it cannot function as a stand-alone computer; it requires external components to be a functional system. The other devices are complete

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